

Judged Aggregation Under Review

A referee's assessment of Freedi's clustering-and-synthesis method and of its comparison with Blair, Procaccia & Tambe's Embeddings for Preferences, Not Semantics

Review of: production pipeline on main · hard-triplet benchmark (875 + pilot-150 replication + fix round) · live accuracy campaign (Findings 1-18) · arXiv 2605.08360 · August 2026

REVIEWER'S SUMMARY

Freedi aggregates free-form statements by dividing one problem between two instruments: embeddings retrieve candidate matches, and a calibrated, fail-closed LLM judgement makes every decision that changes what the public sees. This review asks two questions. What does the evidence establish about that method? And how fairly has it been compared with the strongest alternative — the repaired preference geometry of Blair, Procaccia & Tambe (BPT)?

Three conclusions emerge. First, the method's foundation — that cosine similarity over generic embeddings retrieves well but cannot decide — has been established independently by both groups, with different instruments, and is beyond reasonable doubt. Second, on BPT's own adversarial triplets, judged classification reaches 95–98% where the best repaired geometry reaches 80–81%; this result is correctly measured on production code and survives a model upgrade, yet it is narrower than it looks. It concerns a single-claim pairwise task; its adversaries were machine-generated and judged by a same-family model; and it does not extend to production-scale retrieval, where the strict metric falls to 54–56%. Third, the live pipeline's accuracy arc ($0.067 \rightarrow \approx 0.9$) reflects unusually disciplined measurement, but every fix in that arc was tuned on the same 100-statement corpus that scores it, and no natural-corpus replication yet exists.

The verdicts, claim by claim: the architectural principle is well-supported; the production-accuracy claim is partially supported; the superiority-over-DPT claim holds for the pairwise task only. The review closes with the experiments that would settle what remains open.

1 Scope and materials

Three objects are under review. The first is the Freedi clustering-and-synthesis method as shipped (`functions/src/synthesis/`), described in the companion technical report. The second is the empirical record behind it: the 875-triplet hard-triplet benchmark, its pilot-150 replication with the same-day fix round, the recall-gap probe, and the 100-statement live accuracy campaign in English and Hebrew. The third is the comparison drawn with Blair, Procaccia & Tambe (BPT), *Embeddings for Preferences, Not Semantics* (arXiv 2605.08360), whose released hard triplets the benchmark reuses. The companion report's description of the mechanism was verified against source and is taken as accurate; this review therefore concentrates on a single question — what the evidence does, and does not, establish.

2 The method in one paragraph

A new statement is embedded (OpenAI 3-small/3-large @1536, question-prefixed; long texts are first distilled to briefs). Nearest-neighbour cosine then proposes candidates through calibrated bands, and conjunctive cohesion gates filter them. What remains is decided by a cascade of LLM judgements: a merge requires a four-way "same" verdict and fails closed; synthesis follows a refuse-when-in-doubt rubric with a never-widen-scope rule; themes are filed by their contents, with doubt resolved to NONE; and, optionally, the statement is classified against a growing codebook of canonical claims maintained by conservative consolidation, a broaden-ratchet, and a 5% shadow audit. Scheduled sweeps supply convergence, and splits wait for humans. The whole design compresses to one principle: geometry proposes, judgement disposes.

3 Claim-by-claim assessment

C1 Cosine over generic embeddings is unusable as a merge decision rule.

ESTABLISHED

This claim carries the strongest evidence in either body of work, because the two groups establish it independently and with different instruments. BPT show that base encoders rank a stance-flipped distractor above the true paraphrase 70–95% of the time, and they explain why: a nuisance-dominance decomposition, supported by a theorem. Freedi's benchmark shows the same failure at the extreme — 100.0% of distractors clear the production topical gate, 99.1% clear the strictest merge gate, and 0 of 875 triplets are pipeline-correct — and the production record corroborates it in vivo, in the topic-cluster black hole. A formal analysis and an unrelated production post-mortem here converge on one conclusion. That convergence is what replication is supposed to look like.

C2 LLM-judged placement solves the hard-triplet task (95–98%).

SUPPORTED, WITH THREE QUALIFICATIONS

The measurement itself is sound. The harness imports production functions unmodified; the statistics are appropriate (Wilson intervals, exact McNemar); the results are append-only and resumable; and the finding replicates across a model generation (96.7–98.0% now, 98.7% archived, paired $p = 0.45$).

Three qualifications bound what the number means. **(i) Task shape.** The 95–98% figures come from the single-claim condition, in which match and distractor are classified against exactly the anchor's claim. That condition mirrors BPT's pairwise margin test, so the DPT comparison is fair — but it is not the production task (see C5). **(ii) Family circularity.** The triplets are GPT-4o rewrites, and the classifier is an OpenAI model. A judge reading adversaries written by its own model family may inherit stylistic priors, and those priors would inflate accuracy relative to human-authored paraphrases and stance-flips. BPT are less exposed here, because their scorers are open encoders trained on independently synthesised data. No human-generated hard-triplet set exists to control this — for either system. **(iii) Sample easiness.** On the archived condition, the pilot-150 slice runs about 3.7 points easier than the full 875. Pilot absolutes therefore cannot be quoted against full-set numbers, including BPT's 80.0%.

C3 The live pipeline reaches ≈ 0.90 end-to-end accuracy.

SUPPORTED ON THE BENCHMARK CORPUS; EXTERNAL VALIDITY OPEN

For systems work, the campaign's internal validity is unusually good. Builds were certified byte-for-byte; a deliberate bug-hunt caught two measurement errors that had each inverted a conclusion; arrival orders were seeded; the theme-filing choice followed a pre-registered decision rule; and results are reported as ranges, not best numbers.

The reviewer's concern is adaptive overfitting. The entire fix arc, 0.067 to 0.910, was developed against the same 100-statement corpus that scores it. The seeds vary arrival *order* but never content. The corpus itself is synthetic — 10 topics $\times$ 5 ideas $\times$ 2 paraphrases, uniform in structure, free of off-topic noise and low-effort text — and the Hebrew version, a structure-preserving translation, inherits that same topology plus translationese. Even the striking zero-false-merge result (0 in 450 placements) was measured against pairs that the construction process guaranteed to be mergeable. None of this suggests the numbers are wrong. All of it suggests they are an upper bound. The single most important missing experiment is a frozen-pipeline run on a held-out, naturally collected corpus.

C4

The synthesis layer is exact and order-independent; variance is confined to theming.

SUPPORTED AT N=3 ORDERS, ON-CORPUS

Across three arrival orders, the synthesis layer recovered 150/150 pairs with zero false merges ($P = R = \text{ARI} = 1.000$) — a clean result. The layered-variance argument behind it is also architecturally sound: the layer that changes what people agreed to behaves deterministically, while the layer that merely organises the page absorbs the stochasticity. Two limits remain. Three permutations of one corpus are a small basis for the word "order-independent." And the theming layer is admittedly order-sensitive (topic F1 0.711–0.775, 10–17 headings, recurring attractor headings), so the page a user actually sees is *not* yet order-independent. The report states this honestly; a reader quoting only the synthesis line would not.

C5 The registry beats the paper's best method (95% vs 80%).

SUPPORTED FOR THE PAIRWISE TASK; NOT ESTABLISHED AT PRODUCTION SCALE

On the shared instrument, at the shared task shape, the gap is real and statistically overwhelming (McNemar 827:0 discordant). Three facts temper it.

(i) A standard embedding-gate-plus-judge baseline ties the registry at 95.0% on this corpus, and the adversarial recall-gap probe found the gap between them real but narrow — six Hebrew pairs, 0/6 versus 6/6 — and mostly cross-lingual. The registry's *marginal* value over a well-built gate-and-judge therefore rests on cold-start behaviour, cross-lingual recall, and the maintained codebook: argued convincingly, evidenced thinly. (ii) In the production-shaped setting, where the codebook holds ~94 claims, strict triplet accuracy falls to 36–56%. That strict metric is a lower bound, because the codebook contains near-duplicate claims and 90–93% of matches do attach to *some* legitimate claim; but distractor false-attach also rises to 10–32%, and BPT attempt nothing comparable — so where it matters most, no cross-system number exists at all. The headline "95% vs 80%" describes a laboratory pairwise task, not deployed accuracy. (iii) The fix round exposed model fragility in the enriched, production-shaped variant: a routine upgrade silently cost 12 points through stance-lossy canonicalisation, and the prompt fix recovered only part of it (82.7%), leaving compression loss on multi-part opinions as an inherent limit. The system's own remedy — match on the verbatim anchor, keep the canonical for display — is the right conclusion, and it concedes something worth stating plainly: as a *matching representation*, the canonical-claim layer does not currently beat DPT.

C6 Synthesised texts preserve member intent (fidelity 1.000, 0/98 lost).

PLAUSIBLE; CONSTRUCT NEEDS INDEPENDENT VALIDATION

The scope rule produced a concrete, checkable improvement: scope inflation fell from 4/50 to 0/50. The fidelity score itself is weaker ground. It is an internally defined construct, evaluated by the same model ecosystem that writes the texts, and no human rating study exists. For a system whose central promise is that a synthesis never changes what anyone agreed to, fidelity is the one construct that should not rest on machine self-assessment. A modest blinded study — authors rating whether the merged text still says what they said — would either anchor the claim or reveal what the rubric misses.

C7 Fail-closed engineering contains failures safely.

SUPPORTED, WITH A CAUGHT COUNTEREXAMPLE

The safety taxonomy — placement fails closed, custody fails open, outages are tracked apart from verdicts — is coherent, and the fix round supplied a live demonstration: a deterministic empty-completion failure produced zero wrong merges, exactly the conversion of error into non-action that the design promises. The same episode also shows the residual risk the design cannot remove. Fail-closed turns infrastructure faults into *silent non-decisions*, and the 300-token starvation ran undetected in production until an external benchmark tripped it. The self-audit machinery — the merge count as a recall estimator, the 5% shadow audit — is genuinely original instrumentation; its estimates, however, have not yet been validated against ground truth.

4 Methodological strengths worth naming

- **Benchmarking on production code.** The harness imports the shipped functions, and the campaign's own history shows why this matters: a re-implemented test sweep once measured the copy instead of the code. Most systems papers do not meet this bar.
- **Adversarial reuse of a rival instrument.** Rather than a self-made benchmark, the evaluation uses BPT's released triplets — an instrument designed to break systems of exactly this kind. That choice is the right scientific instinct, and it is what makes the comparison in §5 possible at all.
- **Measurement pathology as a first-class finding.** Two harness bugs that "fail toward plausibility," a review queue with a measured 100% false-positive rate, a run marked "do not cite" after emulator contamination: the campaign treats its own instruments as objects of study. This reviewer wishes the practice were common.
- **Cost-weighted, pre-registered decisions.** The theme-filing variant was chosen by a pre-registered objective (3·misfiles + over-NONEs, with a hard cap), and higher-accuracy variants were explicitly

rejected under it. That is decision-theoretically correct, and rare.

- **Honest range reporting.** "Quote the range, not the best number" — and the enriched-condition regression was published immediately, against the project's own interest. Both lend the record credibility.

5 The comparison itself, refereed

5.1 What is fair

The single-claim registry condition and BPT's triplet margin test share one task shape: score a match and a distractor against a single anchor-derived reference, and count the ordering. Because the task shape is shared and the items are identical, the 875-set numbers — 95.0–96.5% against 80.0–81.1% — constitute a legitimate head-to-head *for that task*, and the McNemar contrast against raw cosine (827:0) is unambiguous. The diagnosis-level agreement of C1 is, likewise, a genuine two-team replication.

5.2 What is not yet fair

AXIS	ASYMMETRY	DIRECTION OF LIKELY BIAS
Inference cost	1–2 LLM calls per placement vs one dot product after encoding; never budget-matched	favours Freedi's accuracy, hides its cost
Family circularity	GPT-generated triplets judged by an OpenAI classifier vs scored by open encoders	plausibly inflates Freedi; magnitude unknown
Pilot-150 replication	easier slice; no per-triplet DPT predictions to pair against, though BPT's code and model are public	inflates pilot absolutes on both sides; paired test absent
Task definition	propositional equivalence vs preferential similarity — siblings that coincide only on stance-flips	neither; limits generalisation of any cross-quote
Deployment task	Freedi's production setting is ~94-way retrieval; BPT report no retrieval-time equivalent	no comparison exists at all where it matters most
Language	Freedi ships bilingual and its Hebrew results bind; BPT is English-only	favours Freedi on scope, untested for BPT

The comparison is real but asymmetric on six axes; a rigorous head-to-head would close the first three.

5.3 The paper, briefly refereed

BPT's contribution stands on its own merits. The preference/nuisance decomposition is a clarifying formal object. Theorem 1 is modest but does real work: it licenses training on synthetic hard triplets without knowing the preference subspace. Training entirely on synthetic counterfactuals and then gaining on 11 natural datasets is a strong generalisation result. And the per-topic projection may be the paper's most consequential finding: a rank-20 linear map, fit on roughly 50 votes over a *frozen* encoder, reaches 81.1% on the hard triplets — which implies the preference signal was in the base embedding all along, merely misweighted. The paper flags its own limits honestly: within-participant rankings only, uncalibrated absolute distances, English-only, and an explicit warning against using such embeddings for binding decisions. What it does not address — because it is not its problem — is everything downstream of geometry: deduplication, labeling, synthesis, opposition structure. Its outputs are inputs to a system like Freedi's, not a substitute for one.

5.4 The reviewer's synthesis

The evidence does not support the reading "judgement beats geometry, 95 to 80." It supports a more precise statement, in three parts. Geometry, even repaired, cannot carry merge decisions alone — that is C1, and it is also BPT's own 80% ceiling on a binary-choice task. Judgement carries those decisions at 95–98%, but at per-decision inference cost, with family-circularity exposure, and only where retrieval has already surfaced the right pair. And the production-scale problem — retrieval among ~100 evolving claims — is convincingly solved by neither, which is exactly where the two would compose: BPT-style preference geometry as the ranker, Freedi-style judgement as the decision rule, and vote-trained per-topic projections learned from Freedi's own evaluation data. Read together, the two records argue for that hybrid more strongly than either argues for itself.

6 Threats to validity, consolidated

- **Internal.** The fix arc adapted to a reused corpus (C3); only three arrival orders were tested; a long campaign accumulated sequential comparisons without correction; and two harness bugs were found — so the prior for a third is not zero.
- **Construct.** The 0.6/0.4 accuracy weighting is a design choice, not a derivation; fidelity is machine-assessed (C6); and the strict multi-claim metric is a lower bound whose distance from truth — the legitimacy of near-duplicate attaches — is itself unmeasured.
- **External.** The corpora are synthetic throughout; the adversarial triplets are machine-generated; only two languages are covered, one by translation; no adversarial participants, spam, or drift appear anywhere; and a single vendor's models perform generation, judgement, and much of the evaluation.
- **Statistical.** At $n = 150$, intervals span ± 4 –7 points — adequate for the 17-point headline gap, inadequate for the 3-point differences the fix round reports as directional.

7 Recommendations

1. **Run DPT on the same items, paired.** BPT's code and model are public. Scoring their encoder on the identical pilot-150 (and the full 875) yields per-triplet McNemar contrasts against the registry — converting the comparison's weakest link, cross-quoted aggregates, into a real paired test, at trivial cost.
2. **Break the family circularity.** Commission a small hard-triplet set written by people — paraphrase and stance-flip authored by hand — and evaluate both systems on it. Only off-distribution adversaries can show whether 95–98% survives them.
3. **Freeze and replicate on a natural corpus.** Take one held-out, naturally collected deliberation (a real Freedi question, with consent), freeze the pipeline before contact, and score pairwise metrics against human annotation. This addresses the central external-validity gap of C3.
4. **Benchmark the task that matters: multi-claim retrieval.** Define a production-shaped benchmark — evolving codebook, near-duplicate-aware scoring — and run three arms: the current registry; gate-and-judge with a DPT or per-topic-projected retriever; and the hybrid. The C-condition numbers say the headroom lives here.
5. **Use the votes.** Freedi already collects the evaluations that BPT's §7 needs. Implement their per-topic rank-r projection on that data as the candidate generator, then measure recall-gap closure and judge-call reduction under a fixed token budget. Of all available experiments, this one offers the most value for the least novelty.
6. **Human-anchor the fidelity construct.** Run blinded author ratings of merged texts, and validate the self-audit estimators against a labeled sample.
7. **Report costs beside accuracy.** Publish tokens and latency per placement for every arm. The architectural argument is strong enough to survive honest cost accounting.
8. **Institutionalise the model-upgrade regression suite.** The fix round showed that a routine upgrade can silently cost 12 points in one link. The pilot-150 harness already exists; wire it to run on any change to `WORKER_MODEL` or the canonicalisation prompt.

8 Verdict

As engineering science, this is careful, self-critical, unusually well-instrumented work. Its central architectural claim — geometry proposes, judgement disposes — draws support from two independent groups and survives the project's own failure catalogue. As a comparative claim against Blair–Procaccia–Tambe, the record supports one precise statement: **on the paper's own adversarial pairwise instrument, judged classification decisively outperforms the best repaired geometry (95–98% vs 80–81%) — at unmeasured relative cost, under a same-family generation confound, and without yet extending to the production-scale retrieval task.** The two approaches are complements with a well-defined composition, and the most valuable next step is not another point on either system's curve. It is the paired, budget-matched, human-adversarial hybrid experiment that neither group has run. The

reviewer's recommendation is the equivalent of *major revision*: as precisely stated, the claims are sound; as they will inevitably be quoted, they need the experiments of §7 behind them.

9 Materials reviewed

- C. Blair, A. D. Procaccia, M. Tambe. *Embeddings for Preferences, Not Semantics*. arXiv:2605.08360, 2026 (full text; code and model releases noted).
- Freedi production source, `functions/src/synthesis/` and `functions/src/services/`, branch `main`.
- Hard-triplet benchmark: `RESULTS.md` (conditions A-D, C/CE/CE2/CH, A2), `RECALL_GAP.md`, `PILOT150-RESULTS.md` incl. the 2026-08-24 fix round.
- Live accuracy campaign: `scientific-research/2026-08-18-live-synth-accuracy/RESULTS.md`, Findings 1-18, certified runs and seed sweep.
- Companion explanation: *Geometry Proposes, Judgement Disposes* (technical report, same series).

This review was prepared against the production source and primary benchmark artifacts, not summaries; where a claim could not be verified from primary materials (e.g. the fidelity scorer's implementation), the review says so rather than assuming. Its prose follows the reader-expectation principles of Gopen & Swan, "The Science of Scientific Writing" (*American Scientist* 78, 1990): context in the topic position, emphasis in the stress position, one point per sentence, subjects near their verbs, actions as verbs. Verdict labels: **ESTABLISHED** = supported by convergent independent evidence; **SUPPORTED WITH QUALIFICATIONS** = correctly measured but bounded in scope; **NOT ESTABLISHED** = no adequate evidence either way.